

- Facilitates rapid scaling and resource optimisation
- A fundamental component of contemporary application deployment

Docker is particularly valuable for teams embracing DevOps practices and aiming to streamline the path from development to production without disruption.

8 Kubernetes: Scaling container orchestration

Kubernetes has become the norm when it comes to the application of containers in the current technology environment because it provides the ability to deploy, monitor, and scale containers. It does this by automating the process of managing containers.

Created to be high-availability and resilient, Kubernetes ensures that even when workloads grow and change, applications can be responsive to users. Its orchestration abilities, in addition to other benefits, make Kubernetes a vital component of large-scale dynamic infrastructure networks.

Kubernetes is indispensable for many reasons:

- Supports deployment, scaling, and management of containers
- Provides high availability solutions for business-critical applications
- Simplifies operations for distributed and cloud-native applications
- Key to organisations dealing with complex workloads

IT teams that want to manage containerised applications at scale without sacrificing operational efficiency are the target audience for Kubernetes.

9 OpenLDAP: Centralised identity and access management

OpenLDAP allows administrators to have a centralised method of controlling user identities and access to multiple servers and applications. Centralised management allows for standardised access policies, thus minimising chances of unauthenticated access. OpenLDAP is commonly employed in enterprises as well as other IT systems that have several users with intricate access issues.

Main strengths of OpenLDAP are:

- Authentication and directory services
- Increases cybersecurity and access control in computer systems
- Simplifies the management of users and permissions
- Widely used for enterprise and hybrid IT environments

OpenLDAP is a necessity for organisations requiring effective identity management as well as access control and compliance.

10 Wireshark: Deep network visibility and troubleshooting

Wireshark is still one of the most trusted tools for network analysis tasks. This is because it enables system as well as

IT administrators to view network traffic for packet-level information to check for problems.

With the ability to do in-depth analysis, Wireshark assists in the troubleshooting of complicated network issues, in addition to analysing bottlenecks and suspicious network communications. Unfortunately, this application originally functioned only on the UNIX operating system.

Wireshark is a must for IT professionals for many reasons:

- Packet-level traffic trace for deep analysis
- Assists in resolving performance and security problems
- Facilitates diagnosis and solution of complex network issues effectively

Wireshark is perfect for an organisation that needs a more detailed analysis of its network for maximum performance and security.

Table 1 gives a quick overview of each category of the IT infrastructure, its function, and the tools essential for it.

Category	Function	Key tools
Monitoring and observability	Provide insights into system performance and help in detecting problems early	Nagios, Zabbix, Prometheus, Grafana
Automation and configuration management	Remove manual labour and ensure uniformity, handling complicated infrastructure with ease	Ansible, Puppet
Containers and orchestration	Deployment, scaling, and lifetime management of containerised applications for cloud-native environments	Docker, Kubernetes
Identity, access and networking	Deploy, scale, and manage applications in containers for cloud native applications.	OpenLDAP, Wireshark

Table 1: The best way system admins can use the top 10 open source tools

These popular open source tools help to monitor systems, automate workflows, and secure infrastructure — all while remaining adaptable against constantly changing challenges.

Empower yourself as a system or IT administrator.

Explore these tools, get them deep into your workflows, and master your ecosystem. Your infrastructure deserves it, and so do your users. Take advantage of open source today and shape the future of your IT operations. **END** 

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